

File permissions in Linux

Project description

In this project, I worked as a security supporting in the research team at a large organization. My responsibility was to ensure that users had the appropriate file permissions in order to keep the system secure. I examined the existing permissions on the file system using Linux commands, compared them against the authorization each user should have, and modified any permissions that didn't match.

Check file and directory details

```
researcher2@690d54f13165:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:40 ..
-rw--w---- 1 researcher2 research_team  46 Jun 22 09:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 22 09:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 22 09:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 22 09:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_t.txt
```

Describe the permissions string

The 10 character string represent the permissions assigned to a file or directory. For example, in -rw-r-r-, the first character (-) indicates that this is a file, not directory. The next three characters (rw-) shows the permissions for the user (owner), who can read and write but not execute the file. The following three characters (rw-) represent the group's permissions, allowing the group to read and write. The final three characters (r-) represent the permissions for the other users, who can also only read the file. In this string, r stands for the read, w for write, x for execute and - means the permissions is not granted.

Change file permissions

```
researcher2@690d54f13165: ~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:40 ..
-rw--w---- 1 researcher2 research_team  46 Jun 22 09:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 22 09:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 22 09:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_t.txt
researcher2@690d54f13165: ~/projects$
```

Removed the write permission on the project_k.txt

Change file permissions on a hidden file

```
researcher2@690d54f13165: ~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:40 ..
-r--r----- 1 researcher2 research_team  46 Jun 22 09:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 22 09:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 22 09:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_t.txt
```

Now, .project_x.txt which is hidden file, have only read permissions to user and group. User cannot write to file anymore.

Change directory permissions

```
researcher2@690d54f13165: ~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:40 ..
-r--r----- 1 researcher2 research_team  46 Jun 22 09:05 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 22 09:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 22 09:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_t.txt
```

Drafts directory now have permissions for execute.

Summary

```
researcher2@690d54f13165:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 22 09:40 ..
-r--r----- 1 researcher2 research_team  46 Jun 22 09:05 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 22 09:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 22 09:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 22 09:05 project_t.txt
researcher2@690d54f13165:~/projects$
```

In this project, we changed the permissions on .project_x.txt which is hidden file, project_k.txt, which removed the write permission, and changed the permissions on drafts directory.